

1. Administrative & Study Identification

Full Study Title: A Phase 3 study of MK-4280A (coformulated favezelimab [MK-4280] plus pembrolizumab [MK-3475]) Versus Standard of Care in Previously Treated Metastatic PD-L1 positive Colorectal Cancer **Short Title/Acronym:** KEYFORM-007 (MK-4280A-007) **Protocol Number:** MK-4280A-007 (EudraCT: 2021-001309-60) **NCT Number:** NCT05064059 **Sponsor Name:** Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc. (Company: Merck) **Investigational Product:** Co-formulated Favezelimab and Pembrolizumab (MK-4280A) **Phase of Development:** Phase 3 **Medical Monitor:** David Fogelman / Merck Clinical Operations

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To evaluate whether the fixed-dose combination of favezelimab and pembrolizumab (MK-4280A) is superior to standard of care (regorafenib or TAS-102) with respect to Overall Survival (OS). * **Secondary Objectives:** To evaluate progression-free survival (PFS), objective response rate (ORR), duration of response (DOR), safety, and quality of life.

2.2 Background & Rationale Effective treatment options remain a critical unmet need for patients with previously treated metastatic colorectal cancer (mCRC), especially those who have microsatellite stable (MSS) / mismatch repair proficient (pMMR) disease, which historically has limited response to immunotherapies. Blockade of both the lymphocyte-activation gene 3 (LAG-3) and programmed cell death protein 1 (PD-1) pathways using favezelimab and pembrolizumab showed promising antitumor activity in early studies. This study aims to compare this combination against standard chemotherapy regimens in PD-L1 positive MSS mCRC.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Active Comparator (Standard of Care: Regorafenib or TAS-102) * **Blinding:** Open-label * **Randomization:** 1:1

3.2 Planned Enrollment & Duration * **Target \$N\$:** 441 participants * **Number of Sites:** Multi-site (approximately 21 countries) * **Trial Status:** Completed * **Estimated Study Start:** Late 2021 * **Estimated Primary Completion:** Final analysis conducted (trial did not meet primary endpoint as reported in Sept 2024)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Has a histologically confirmed colorectal adenocarcinoma that is metastatic and unresectable. 2. Has measurable disease per RECIST 1.1 as assessed by the local site investigator. 3. Has been previously treated for the disease and radiographically progressed on or after (or could not tolerate) standard treatment. 4. PD-L1 positive and microsatellite stable (MSS) / mismatch repair proficient (pMMR) status. 5. Submits an archival (≤ 5 years) or newly obtained tumor tissue sample that has not been previously irradiated. 6. Has an Eastern Cooperative Oncology Group (ECOG) Performance Score of 0 to 1 within 10 days prior to first dose of study intervention. 7. Has a life expectancy of at least 3 months, based on the investigator assessment. 8. Has the ability to swallow and retain oral medication and not have any clinically significant gastrointestinal abnormalities that might alter absorption. 9. Has adequate organ function.

4.2 Exclusion Criteria 1. Has previously been found to have deficient mismatch repair/microsatellite instability-high (dMMR/MSI-H) tumor status. 2. Has known active central nervous system (CNS) metastases and/or carcinomatous meningitis or leptomeningeal disease. 3. Has a history of acute or chronic pancreatitis. 4. Has neuromuscular disorders associated with an elevated creatine kinase (e.g., inflammatory myopathies, muscular dystrophy, amyotrophic lateral sclerosis, spinal muscular atrophy). 5. Has clinically significant cardiovascular disease within 12 months from first dose of study intervention. 6. Has urine protein greater than or equal to 1g/24h. 7. A woman of childbearing potential who has a positive urine/serum pregnancy test within 24/72 hours prior to the first dose of study intervention. 8. Has received prior therapy with an anti-PD-1, anti-PD-L1, anti-PD-L2, anti-LAG-3 antibody, or with a tyrosine kinase inhibitor (other than RAF inhibitors), or an agent directed to another stimulatory or coinhibitory T-cell receptor. 9. Has previously received regorafenib or TAS-102. 10. Has received prior systemic anticancer therapy including investigational agents within 28 days before randomization. 11. Has received prior radiotherapy within 2 weeks of start of study intervention. 12. Has received a live or live-attenuated vaccine within 30 days before the first dose of study intervention. 13. Has a known additional malignancy that is progressing or has required active treatment within the past 3 years. 14. Has severe hypersensitivity ($\geq$ Grade 3) to pembrolizumab and/or any of its excipients. 15. Has an active autoimmune disease that has required systemic treatment in past 2 years. 16. Has a history of (noninfectious) pneumonitis/interstitial lung disease that required steroids or has current pneumonitis/interstitial lung disease. 17. Has an active infection requiring systemic therapy (e.g., tuberculosis, known viral or bacterial infections), or a known history of HIV, Hepatitis B, or active Hepatitis C virus

infection. 18. Has a history or current evidence of any condition, therapy, or laboratory abnormality that might confound the results of the study. 19. Has had an allogenic tissue/solid organ transplant.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** Arm A: Fixed-dose combination of favezelimab (800 mg) and pembrolizumab (200 mg) administered on Day 1 of each 3-week cycle. * **Route of Administration:** Intravenous (IV). * **Duration of Treatment:** Up to 35 cycles (approximately 2 years) or until disease progression/unacceptable toxicity. * **Comparator Arm:** Arm B: Investigator's choice of regorafenib (oral, 4-week cycles) [Trade name: Stivarga, ATC Code: L01EX05] or TAS-102/trifluridine and tipiracil (oral, 4-week cycles).

5.2 Concomitant & Prohibited Therapy * **Permitted:** Standard supportive care for symptoms associated with mCRC. * **Prohibited:** Concurrent anti-cancer therapies, investigational agents, or live vaccines during the trial.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor tissue submission, Imaging
Baseline	Day 0	Randomization, First Dose, Vitals, Baseline Labs
Treatment Cycles	Q3W or Q4W	Drug administration, Safety Labs, Efficacy Assessment (Imaging), AE monitoring
End of Study		Follow-up Survival tracking, Exit Interview, Final Radiographic Scans

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Overall Survival (OS) - Defined as the time from randomization to death due to any cause. * **Measurement Method:** Clinical tracking and survival follow-up assessments.

7.2 Secondary & Exploratory Endpoints * Progression-Free Survival (PFS) per RECIST 1.1 as Assessed by Blinded Independent Central Review (BICR) * Objective Response Rate (ORR) per RECIST 1.1 as Assessed by BICR * Duration of Response (DOR) per RECIST 1.1 as Assessed by BICR * Incidence of Treatment-Related Adverse Events (TRAEs)

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Continuous monitoring at each cycle visit (evaluating Treatment-Related Adverse Events and Immune-Mediated Adverse Events).
 - **Serious Adverse Events (SAEs):** Standard reporting timeline (typically 24 hours to the sponsor and Regulatory Authorities).
 - **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee (iDMC) utilized for periodic safety and efficacy oversight.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) population for all efficacy endpoints (OS, PFS, ORR). Safety Set for patients receiving at least one dose of study drug.
 - **Sample Size Justification:** \$N=441\$ powered to detect a statistically significant superiority in OS for the MK-4280A arm vs. standard of care.
 - **Handling of Missing Data:** Standard Kaplan-Meier methodologies with right-censoring for survival endpoints.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Onsite and remote monitoring visits periodically throughout the enrollment and treatment phases.
 - **Audit Trail:** Standard EDC (Electronic Data Capture) systems utilizing eCRF locking and query resolution.
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11. Study Identifiers & Governance

- **Primary ID:** MK-4280A-007
 - **Posting ID:** 433584783679003202
 - **Registry Number:** NCT05064059
 - **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC
 - **Study Title:** KEYFORM-007
 - **Official Regulatory Title:** A Phase 3 study of MK-4280A (coformulated favezelimab [MK-4280] plus pembrolizumab [MK-3475]) Versus Standard of Care in Previously Treated Metastatic PD-L1 positive Colorectal Cancer
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12. Clinical Framework (Colorectal Cancer Specific)

- **Primary Condition:** Metastatic Colorectal Cancer (mCRC)

- **UMLS Preferred Name:** Metastatic Colorectal Cancer (Neoplastic Process)
 - **MeSH Code:** D015179
 - **SNOMED ID:** 363406005
 - **Diagnosis Threshold:** PD-L1 positive, microsatellite stable (MSS) / mismatch repair proficient (pMMR)
 - **Medical Methodology:** Immunotherapy (Anti-LAG-3 + Anti-PD-1) vs. Standard Chemotherapy
 - **Optimization Target:** OS and PFS improvement in refractory settings
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13. Study Procedures & Methodology

- **Management Pathway:** Intravenous investigational immunotherapy vs standard-of-care oral agents
 - **Education Protocol (HCP):** Site initiation visits, protocol training, and immune-mediated AE management guidelines
 - **Education Protocol (Patient):** Informed consent process, dosing diaries (for oral comparators), and side-effect reporting education
 - **Quality Audit Mechanism:** Routine source data verification and clinical site monitoring
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14. Observation & Follow-Up Schedule

- **Total Duration:** Up to 35 treatment cycles (~2 years) plus survival follow-up
 - **Onsite Visit Cadence:** Every 3 weeks (Arm A) or Every 4 weeks (Arm B)
 - **Remote Monitoring Frequency:** As indicated for survival status follow-up post-treatment
 - **Checklist Review Interval:** Prior to the administration of each cycle
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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Insert PI Name]	_____	06-Apr-2026
Clinical Operations Lead	David Fogelman	_____	06-Apr-2026
Lead Statistician	[Insert Statistician Name]	_____	06-Apr-2026